**SDC HACKATHON 2026**

# Product Requirement Document (Updated for the Final Round)

Future Skills · Team Hands Off Our Rights · Domain 1: Aiversecity Future-Ready Workforce

*Revision of "SDC2026_HandsOffOurRights_Preliminary.pdf" (25 October 2026). Updated 1 November 2026.*

## 0. What Changed Since the Preliminary Submission, and Why

The Final Round Finalist Guide raised the bar: all six must-haves must run live, judges may change the trend data mid-demo, check that a course is real, and ask why a score is what it is. What we learned from the data and from testing the flow shaped the rest.

| Area | Preliminary (25 Oct) | Final (1 Nov) | Why |
|---|---|---|---|
| IT areas and roles | 6 areas, 26 roles | 5 areas, 25 roles: IT Management dropped; Statisticians and Special Effects Artists and Animators added | IT Management roles had no usable Malaysian demand signal; the two added roles widen the data and creative options for students |
| Trend score | Mostly DOSM online job ads | v3: Malaysian demand = share of an area's roles on MyCOL 2024/2025 (+ DOSM where it exists); AI momentum = continuous arXiv growth scale | DOSM publishes job ads for only one IT occupation, so the old score barely separated areas; MyCOL is official and per-occupation |
| Change test | Update data button only | "What if the trend data changed?" page: change any input and see signals → scores → ranking before and after | Judges will ask "change the trend data now, what changes?" |
| Trend radar | One page among many | Landing page, with source and date links for every input, and Update data at the top | Must-have 2 requires source and date on screen |
| Learning pathway | 26 of 186 needed skills had a free checked course (14%) | 156 of 191 (81.7%); every job has its own resources; link checker; progress tracking without accounts | Phase 1 KPI 2 target was 40%; judges may check a course is real |
| Matches page | Long cards | Scenario switch on top, SDG icons and score bars on every card, short AI "Why this job?" | The scenario and the score parts should be seen first; the guide says SDG tags must change something for the user |
| Languages and access | English | English, Bahasa Melayu, Chinese; light/dark mode; gzip and ETags; 40 px touch targets | Good-to-have "inclusive design" and low-bandwidth use |
| LLM | gpt-oss-120b on Fireworks AI (team's own account) | gpt-5.4-mini through the SDC sponsor KrackedDevs gateway; any OpenAI-compatible API works; same guardrails | Use the sponsor resource and avoid single-provider risk |
| Trust | — | Score ranges on the skills test, exam-section links per question, CV privacy box, progress bar while the CV is read | A 3-question area score is uncertain and a CV is personal data, so the app says so plainly |

## 1. Team & Project Overview

| | |
|---|---|
| Team Name | Hands Off Our Rights |
| Selected Domain | Domain 1: Aiversecity Future-Ready Workforce |
| SDG Track | SDG 4 Quality Education (4.3, 4.4); SDG 8 Decent Work and Economic Growth (8.2, 8.5) |
| Project / Solution Name | Future Skills |
| Name (Leader) | Hong Zhi Yang |
| Email & Phone (Leader) | zhiyang19015@gmail.com \| +60 11-4035 9109 |
| University Affiliation | Universiti Tunku Abdul Rahman (UTAR) |
| Number of Team Members | 3 |
| Live Prototype | https://future-skills.replit.app |
| Code Repository | https://github.com/pekoremon/future-skills |

## 2. Problem Statement

**2.1 Problem context.** Malaysia's graduates often end up in jobs that do not use their education: in Q2 2026, 35.2% of employed people with tertiary education (1.95 million) worked in semi-skilled or low-skilled jobs (DOSM, 2026b). Youth unemployment (15–24) was 10.3% in 2024, against 3.2% nationally (DOSM, 2026a). AI is spreading faster than training: 56% of Malaysian professionals use AI tools at work but only 26% have had formal training (Hays Asia Salary Guide 2025, quoted in the Domain 1 brief).

We focus on IT and computing undergraduates choosing a specialisation, an internship or a first job. “IT” is really five different areas (software engineering, data and AI, cybersecurity, cloud and networking, web and interface design), each needing different interests and skills, and AI is changing which roles are in demand faster than course guides are updated. The information exists (TalentCorp MyCOL, DOSM job-ad and wage data, free courses) but is scattered and not linked to the individual. No tool connects it to one student's interests and skills, or turns a skills gap into a plan.

**2.2 Target users.** Primary: IT and computing undergraduates in Malaysian universities, polytechnics and TVET colleges. Secondary: career centres advising them, and working adults reskilling into IT (HRD Corp-claimable courses are listed for them).

## 3. Solution and Key Features

Future Skills is a web app that shows an IT student which IT area suits them, which of 25 IT roles fit, which skills they lack, and a free plan to learn them, with Malaysian demand and AI-trend signals for every role. **GenAI explains; our data decides.**

| Must-have (Domain 1) | What the student sees in the final build |
|---|---|
| 1. Skills profile intake | Optional CV (PDF/DOCX/text) or 1–5 skill ratings → 36-item interest questionnaire (RIASEC + IT area) → skills test on only the 3–5 areas these point to (15–21 questions, hands-on SQL and Python), with score ranges |
| 2. AI-trend signal layer | Future Skills Radar landing page: MyCOL, DOSM/OpenDOSM and arXiv inputs with source and date, direction and confidence per area and role; “Update data” refreshes live; “What if” lets anyone change an input |
| 3. Future-role matching | 25 roles ranked by 0.4 area fit + 0.1 interest + 0.3 skill + 0.2 trend; every part shown, plus the missing skills and what would raise the score |
| 4. Personalised reskilling pathway | Gaps ordered biggest first, checked free courses and videos with hours and weeks at the student's study time, HRD Corp-claimable options listed apart, certificates and project ideas, progress tracking |
| 5. SDG mapping | Every role and skill tagged to SDG targets with a one-line reason (233 links); the path sums up which targets it contributes to |
| 6. Handling uncertainty | Three scenarios (current trends, fast AI adoption, Malaysian demand leads) re-weight measured trend parts; rankings shift and are labelled as estimates |

**How AI is applied.** (1) CV reading: an LLM (gpt-5.4-mini via KrackedDevs) quotes CV evidence per IT area; our code keeps only quotes really in the CV and scores them. The CV is never stored. (2) “Why this job?”: the LLM turns one match's computed facts into plain language; a reply with any number not in those facts is rejected and a rule-based summary shown. (3) Matching, trends, scenarios and pathways are explainable code over sourced tables.

## 4. Scope and Boundaries

| In scope | Out of scope |
|---|---|
| IT career choice for students: 5 IT areas, 25 roles, Malaysian demand data, free learning paths, SDG links, scenarios; English, BM and Chinese; phone and desktop | Non-IT careers; writing course content; payments; real recruitment integrations or hiring decisions (the student decides); storing CVs or personal data; real CVs in demos |

## 5. Technical Architecture

| Layer | Technology |
|---|---|
| Frontend | HTML, CSS, plain JavaScript (no framework, no build step); PDF.js and Pyodide in the browser; EN/BM/中文 |
| Backend | Python 3.12 standard library: JSON API on http.server, gzip + ETags; no third-party packages |
| Data | CSV/JSON tables (roles, skills, role–skill links, courses, resources, SDG map, trend inputs/signals/scores); every row has source name, URL and date; validate_data.py and check_links.py |
| GenAI | Any OpenAI-compatible chat API; live: gpt-5.4-mini via the SDC sponsor KrackedDevs gateway, with output checks and rule-based fallback |
| External data APIs | OpenDOSM API, arXiv API (live refresh); Microsoft Learn Catalog API (development only) |
| Hosting / Dev | Replit Workspace and Deployments (SDC sponsor), live at https://future-skills.replit.app; GitHub; KrackedDevs GenAI keys (SDC sponsor) |
| Testing | 172 unittest tests, data validator, link checker, kpi.py |

## 6. SDG Impact Framework and KPIs

| Item | Phase 1 definition | Status at the Final (1 Nov 2026) |
|---|---|---|
| Target SDG sub-targets | 4.4 relevant skills for decent jobs; supporting 8.5 full and productive employment | Unchanged; 4.3 and 8.2 also tagged where a role or skill supports them |
| Problem baseline | 35.2% of tertiary-educated workers in semi/low-skilled jobs (DOSM Q2 2026); 26 of 186 skills (14%) with a checked course | Same national baseline; prototype baseline re-measured with scripts/kpi.py at the Phase 1 commit 6802b63: 54 of 186 (29.0%) counting free videos too |
| KPI 1: Career clarity | ≥70% of ≥20 IT student testers can name a target role, its top missing skills and a next course after one session | Pilot at the physical hackathon: **[TO FILL: N testers; X% before → Y% after]** |
| KPI 2: Pathway coverage | Raise checked free-course coverage to 40% (about 75 skills) | 81.7% (156 of 191 needed skills), measured by python scripts/kpi.py: target exceeded |

## 7. Design Thinking, Deployment and Next Steps

**Empathise / Define:** we are IT students facing the same choice; students need roles they had not considered, proof that a trend is real, and a concrete next step. **Ideate:** transparent scoring over a black box, and one rule: never invent a number. **Prototype and test:** students gave up on the full 24-question test, so the flow became CV → interests → a focused test; the radar moved to the landing page because judges and students both start with "what is changing?". **Deploy:** the app is live on Replit, needs no packages, and still works without the LLM key or when a data source fails (the last good data is kept). **Next:** run the career-clarity study with partner career centres, add each new DOSM Job Market Insights quarter and MyCOL edition, skill-level signals from job ads, an HR team view, and a fairness test.

## 8. Sources


• DOSM (2026a). Labour Market Review, Q1 2026. https://storage.dosm.gov.my/analysis/lmr_2026-q1_en.pdf

• DOSM (2026b). Labour Market Review, Q2 2026. https://storage.dosm.gov.my/analysis/lmr_2026-q2_en.pdf

• TalentCorp (2025). Malaysia Critical Occupations List (MyCOL) 2024/2025. https://www.talentcorp.com.my/mycol/

• Hays (2025). Hays Asia Salary Guide 2025, as quoted in the SDC 2026 Domain 1 brief.

• United Nations. Global indicator framework for the SDGs (A/RES/71/313). https://unstats.un.org/sdgs/indicators/indicators-list/

• Future Skills team (2026). Data tables and SOURCES.md. https://github.com/pekoremon/future-skills